Pricing Derivatives and the Derivation of the Black-Scholes-Merton PDE

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Introduction to Derivatives

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Derivatives are financial contracts whose value is derived from an underlying asset, such as stocks, bonds, commodities, or currencies. Examples of derivatives are options, future contracts, forwards, and swaps.

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Definition (Options)

Options contracts provide the buyer or investor with the right, but not the obligation, to buy or sell an underlying asset at a preset price, called the strike price K, at time T, the expiration time.

Introduction to Options

Example (Call options: Rare comic book)

- Options $\approx$ insurance
  - Call options: buy
  - Put options: sell
- a friend owns a rare comic book ($90 at time t)
- you believe its value will increase
- you want to profit from it and at the same time avoid losing money if the comic book's value goes down
- Buy a call option ($ 10)
  - allows you to buy the comic book at time T with $ 100 no matter what its market value is

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- **Options trading: a multi-billion-dollar market**
  - Question: Average daily value traded in single-stock options in the US?
  - $432 billion (Wall Street Journal)
  - about 108,000 yachts
- Maximize profits
- Making informed decisions on when to trade an option

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- Provides a reference for the determination of the price of an option using a structured methodology
- Led to a boom in options trading

$$\boxed{V_t = rV - rSV_s - \frac{\sigma^2}{2} S^2 V_{ss}}$$

Brownian Motion

Definition (Brownian Motion)

Let $(\Omega, \mathcal{F}, \mathcal{P})$ be a probability space. A *Brownian motion* $\{W_t\}_{t \geq 0}$ is a family of random variables $W(t) : \Omega \to \mathbb{R}$ satisfying the properties:

1. $W_0 = 0$.
2. **(Continuous Path)** For each $\omega \in \Omega$ and for all $t \in [0, \infty)$, the path $t \to W_t(\omega)$ is continuous with probability one.
3. **(Normally Distributed Increments)** $W_t - W_s$ is normally distributed with mean zero and variance $t - s$, for $t > s \geq 0$.
4. **(Independent Increments)** Brownian motion has *independent increments*. That is, for all $0 \leq t_0 < t_1 < ... < t_n$, the increments

$$W_{t_1} - W_{t_0}, \; W_{t_2} - W_{t_1}, \; ..., \; W_{t_n} - W_{t_{n-1}},$$

are independent random variables.

- We will use W_t to model randomness of the market.

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- **(Calc 1010 - 10 times harder)**

$$\underbrace{dy}_{\text{change in y}} = \underbrace{udt}_{\text{u/change in t}} + \underbrace{vdW_t}_{\text{v/change in Brownian motion}}$$

Ito's Lemma

Theorem (Ito's Lemma)

Suppose we are given the stochastic differential equation

$$dX_t = udt + vdW_t$$

and a smooth function $f(t, x)$. Then the stochastic process $Y_t = f(t, X_t)$ can be written in the differential form

$$dY_t = f_t(t, X_t)dt + f_x(t, X_t)dX_t + \frac{1}{2}f_{xx}(t, X_t)(dX_t)^2.$$

Formal Rules: $(dt^2) = 0 = dtdW_t = dW_t dt$ *and* $(dW_t)^2 = dt$.

- Ito's Lemma act as the "chain rule" in stochastic differential equation.

Ito's Lemma: Simple Example

Example

Consider the stochastic process $f(t, W_t) = tW_t$. We need to find an expression for $d(tW_t)$.
We have:

$$f_t = W_t, f_{W_t} = t, f_{W_t W_t} = 0$$

Applying Ito's lemma, we have:

$$\begin{aligned} df(t, W_t) &= f_t(t, W_t)dt + f_{W_t}(t, W_t)dW_t + \frac{1}{2} f_{W_t W_t}(t, W_t)(dW_t)^2 \\ &= W_t dt + t dW_t + 0 = W_t dt + t dW_t. \end{aligned}$$

Thus, $d(tW_t) = W_t dt + t dW_t$.

The No-Arbitrage Principle

Definition (The No-Arbitrage Principle)

There should be no possibility of exploiting market inefficiencies or price discrepancies to generate guaranteed profits. Basically, it means if two assets have same future value, they also have same present value.
If $\forall\ \omega \in \Omega,\ X(T,\omega) = V(T,\omega)$, then $X(t) = V(t)\ \forall\ 0 \leq t \leq T$.

Example (Arbitrage opportunity)

- At time t, let there be 2 assets V and X such that $X(t) < V(t)$ and $X(T,\omega) = V(T,\omega)\ \forall\ \omega \in \Omega$.
- Person A owns $V(t)$. I borrow $V(t)$ from person A, then sell it to person B.
- At the same time t, buy $X(t)$ from person C.
- At time $T > t$, I return $X(T)$ to A.

BSM Derivation: Assumptions and Notation

1. X: Portfolio (Stock and bond)
   S: Stock price
   Δ: Shares of stock
   r: interest rate of bond (risk-free return)

$$dX = \underbrace{r(X - \Delta S)}_{Bonds\ in\ X} dt + \Delta dS$$

2. Stock price change follows Geometric Brownian motion

$$dS = \underbrace{\alpha S dt}_{drift\ term} + \underbrace{\sigma S dW_t}_{stochastic\ term}$$

   - α : average rate of return on the stock
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3. **Goal**: Find the value of $V(t)$ at time $t = 0$, with T being the terminal time.

Derivation of the BSM Equation

$$dX = r(X - \Delta S)dt + \Delta dS = r(X - \Delta S)dt + \Delta \underbrace{(\alpha S dt + \sigma S dW_t)}_{\text{Geometric Brownian Motion}}$$

$$= rXdt + (\alpha - r)\Delta S dt + \sigma \Delta S dW_t$$

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Construct portfolio X such that $X(T, \omega) = V(T, \omega) \; \forall \; \omega \in \Omega$

No-arbitrage principle $\rightarrow X(t) = V(t) \; \forall \; 0 \leq t \leq T$

$$dX = dV$$

We want to equate dX and dV, so we apply Ito's lemma to dV and get

$$dV = V_t dt + V_s dS + \frac{1}{2} V_{ss} (dS)^2$$

Derivation of the BSM Equation Cont'd

Equating

$$dX = rXdt + (\alpha - r)\Delta Sdt + \sigma \Delta SdW_t$$

and

$$dV = V_t dt + V_s dS + \frac{1}{2} V_{ss} (dS)^2,$$

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then substiting in $dS = \alpha Sdt + \sigma SdW_t$, we get

$$\begin{aligned} & rVdt + (\alpha - r)\Delta Sdt + \sigma \Delta SdW_t \\ = & V_t dt + V_s(\alpha Sdt + \sigma SdW_t) + \frac{1}{2} V_s(\sigma^2 S^2 (dW_t)^2). \end{aligned}$$

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Applying the rule from Ito's lemma $(dW_t)^2 = dt$

$$rVdt + (\alpha - r)\Delta Sdt + \sigma \Delta SdW_t = V_t dt + V_s(\alpha Sdt + \sigma SdW_t) + \frac{1}{2} V_s(\sigma^2 S^2 dt)$$

Derivation of the BSM Equation Cont'd

Let $\Delta = V_s$ and multiply out terms,

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$$-V_t = -rV + rSV_s + \frac{\sigma^2}{2}S^2 V_{ss}$$

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$$\boxed{V_t = rV - rSV_s - \frac{\sigma^2}{2}S^2 V_{ss}}$$

BSM Initial Boundary Value problem

- $V(S(T), T)$ is the value of the option at time T given stock price $S(T)$.

- We assume this is a function of $S(T)$. Take the call options for example.

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Closed-form Solution of the European Call options

$$\Rightarrow V(S,t) = S\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_-} e^{-(y+\sigma\sqrt{T-t})^2/2}dy$$
$$- Ke^{-r(T-t)}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_-} e^{-y^2/2}dy,$$

with $d_- = \frac{1}{\sigma\sqrt{T-t}}\left(\ln\left(\frac{S}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)$

Finite difference method

The finite difference method is a way to find the solution to a initial boundary value problem (IBVP) by using estimation of the derivatives. Let Δt and ΔS be the change in time t and the change in stock price S. Let $V_{n,j} = V(n, T - j\Delta t)$.

$$\frac{\partial V}{\partial t} \approx \frac{V_{n,j+1} - V_{n,j}}{\Delta t}$$
$$\frac{\partial V}{\partial S} \approx \frac{V_{n+1,j} - V_{n-1,j}}{2\Delta S}$$
$$\frac{\partial^2 V}{\partial S^2} \approx \frac{V_{n+1,j} - 2Vn,j + V_{n-1,j}}{\Delta S^2}$$

Finite difference method

Plug in:

$$rV_{n,j} \approx \frac{V_{n,j+1} - V_{n,j}}{\Delta t} + rS\left(\frac{V_{n+1,j} - V_{n-1,j}}{2\Delta S}\right) + \frac{1}{2}\sigma^2 S(t)^2\left(\frac{V_{n+1,j} - 2Vn,j + V_{n-1,j}}{\Delta S^2}\right)$$

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$$\Rightarrow V_{n,j+1} = V_{n,j} + \Delta t(rV_{n,j} - rS(\frac{V_{n+1,j} - V_{n-1,j}}{2\Delta S}) - \frac{1}{2}\sigma^2 S^2(\frac{V_{n+1,j} - 2V_{n,j} + V_{n-1,j}}{\Delta S^2}))$$

Graph for Call options

Let strike price $K = 5$, average variance of stock price $\sigma = 0.2$, and risk-free interest rate $r = 0.15$. The expiration time is $\mathrm{T} = 1$, and the stock price $S = [0, 10]$.

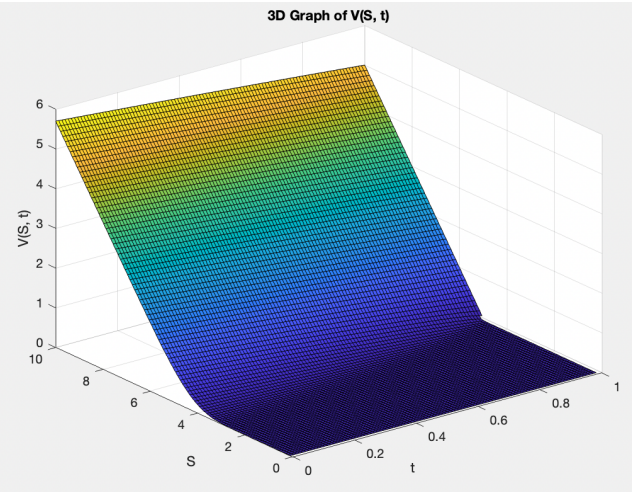

Figure: Call options graph using closed-form formula

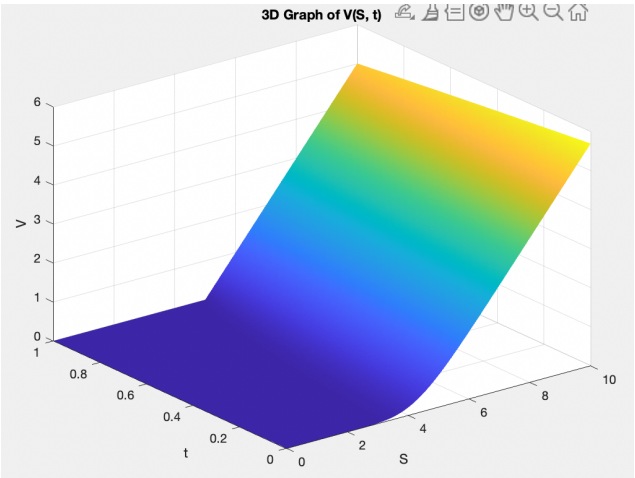

Figure: Call options graph using finite difference method

Future Directions

- **Research Objectives**: build a more robust model that take accounts of changes in equity premium and variance of stocks as well as interest rate of the bond.

- Use finite difference method to estimate the price of options.

Cox-Ingersoll-Ross Interest Rate Model

The Cox–Ingersoll–Ross (CIR) model describes the changes of interest rates:

$$dr_t = a(b - r_t)dt + \sigma\sqrt{r_t}dW_t$$

Heston Model Time-varying Variance Model

Following the Cox-Ingersoll-Ross (CIR) model (1985), Heston's time-varying variance model (1993) can be used to take account of changes in variance of stocks:

$$\begin{cases} dS(t) = \mu S(t)dt + \sqrt{\sigma_s(t)}S(t)dW_1(t) \\ \\ d\sigma_s(t) = \kappa(\bar{\sigma} - \sigma_s(t))dt + \eta\sqrt{\sigma_s(t)}\left(\rho dW_1(t) + \sqrt{1-\rho}dW_2(t)\right). \end{cases}$$

Campbell and Viceira Time-varying Premium Model

As in Campbell and Viceira (1999, 2002), a time-varying equity premium model can be used to take account of changes in equity premium of stocks:

$$\begin{cases} dS(t) = (\mu + X(t) + r)S(t)dt + \sigma_s S(t)dW_1(t) \\ \\ dX(t) = -\kappa X(t)dt + \sigma_X \left(\rho dW_1(t) + \sqrt{1-\rho}dW_2(t)\right). \end{cases}$$

(*) Equity premium is the difference between the return on a stock and the return on a bond.

New Model

Combining mentioned models, the model we will consider is

$$\begin{cases} dS(t) = (\mu + X(t) + R(t))S(t)dt + \sqrt{\sigma_s(t)}S(t)dW_1(t) \\ dX(t) = -\kappa_X X(t)dt + \sigma_X \left(\rho_X dW_1(t) + \sqrt{1-\rho_X}dW_2(t)\right) \\ d\sigma_s(t) = \kappa_s(\sigma - \sigma_s(t))dt + \eta\sqrt{\sigma_s(t)}\left(\rho_s dW_1(t) + \sqrt{1-\rho_s}dW_3(t)\right) \\ dR(t) = \kappa_r(r - R(t))dt + \xi\left(\rho_r dW_1(t) + \sqrt{1-\rho_r}dW_4(t)\right) \end{cases}$$

New Model

Parameters:

- r is the risk-free rate
- μ is the long term average equity premium on stocks
- η is the volatility of the volatility
- ρ_X, ρ_s, and ρ_r is the correlation between the equity premium and the stock price, between W_1 and W_3, and between W_1 and W_4
- σ, σ_X, and σ_s is the long term average variance on the stock, the variance of the equity premium, and the variance of the variance
- κ_X, κ_s, and κ_r is the pressure for X, for the variance, and for the interest rate to return to its average

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